Pranav Sharma

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I'm a creative programmer in pursuit of a deeper sense of fulfillment in my work. I'm passionate about robotics and computer vision and have industry and academic experience in the fields. I aspire to see and interact with my work in an interesting, cutting-edge product which may benefit society in an impactful way. I hope to be afforded that opportunity at a company that aligns with my ideals for a prosperous future. More about me and my work can be found at my website.

EDUCATION | University of Central Florida

Computer Science BS | 3.9 GPA

Mathematics Minor, Intelligent Robotic Systems Minor

Computer Science Skills

Languages: C++, MATLAB, C#, Java, Python, JavaScript, HTML, CSS

Software: Git, ROS, OpenCV, PlatformIO, Gazebo, Unity, Unreal Enginec, CMake

Hobbies

3D Modeling & Printing, Embroidery, Sewing, Sculpting,, Snowboarding, Scuba Diving

Academic Achievements

2020: UCF Magna Cum Laude

2017: UCF Provost Scholar, HS Salutatorian

EXPERIENCE

BALYO

Robotics Field Engineer | June 2025 - Present

- Work closely with stakeholders at customer sites to deploy and troubleshoot automated forklift systems
- Integrate sensors like LIDAR and encoders at the hardware interface level for real-time data processing
- Debug algorithms such as SLAM, path planning, and controls within real-time embedded applications
- Analyze requirements and system design for embedded software, then integrate, test, and validate
- Diagnose and repair issues with mechanical and electrical systems

MICROSOFT

Contract Software Engineer / Team Lead | Windows Servicing & Delivery | January 2024 - May 2025

- Liaise with stakeholders and manage responsibilities and priorities of the vendor team
- Write, test, and deploy bug fixes, and comprehensively test new feature development
- Debug the C/C++ codebase to solve issues customer-facing issues

Software Engineer | Windows Wi-Fi | July 2020 - May 2023

- Onboarded new test lab, developed C#-based tools and automation utilities to improve testing efficiency
- Refactored C/C++ codebase with modern constructs and compile with proprietary build system
- Diagnosed and debugged issues found by customer bug reports, security consultants, and static analysis

LOCKHEED MARTIN

UCF Student Software Engineer Contractor | Embedded Systems | Sept 2019 - January 2020

- Built graphical tool to parse, visualize, and analyze sensor telemetry from CSV data using Matlab
- Simulated and benchmarked asynchronous local data transmission between systems using C++

Software Engineer Intern | Modeling & Simulation | June 2019 - August 2019

- Upgraded C++ simulation framework to utilize config files for parameters, implemented multithreading

UCF Student Software Engineer Contractor | Applied Research | Sept 2018 - May 2019

- Built, calibrated, and tested drone systems with radio, flight controller, and NVIDIA AI vision system in the lab and field, using OpenCV, ArduPilot, and C++ code, compiled with CMake build system

PROJECTS

Embedded Software, Computer Vision | Hydroponic System | May 2024 - March 2025

- Developed low-level embedded C program to capture images with a microcontroller and camera
- Developed Python scripts with OpenCV to automate data collection and analysis

Embedded Software | Robot Arm | May 2024

- Developed embedded C++ program for microcontroller and hardware interfaces for robotic system
- Calculated arm's inverse kinematics so the end could move to specific points in space to draw pictures

Robotics Software Development | UCF CS Senior Design | Spring 2020

- Created an autonomous lawnmower working alongside electrical and mechanical engineering teams
- Developed and tested SLAM and pathing algorithms with LIDAR and camera in simulation using the Robot Operating System and Gazebo, deployed on an NVIDIA Jetson Nano-powered robot, written in Python